Experiment No-28 : Medical Diagnosis

Aim:

To write a Prolog program for Medical Diagnosis that identifies a disease from the symptoms of a patient.

Programming Code:

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% NOTE: For academic demonstration only. Real diagnosis must be made by a doctor.

% ---------- Facts ----------
patient(ravi).
patient(priya).
patient(arun).
patient(divya).
patient(kiran).
patient(sam).

% patient_symptom(Patient, Symptom)
patient_symptom(ravi, fever).
patient_symptom(ravi, cough).
patient_symptom(ravi, body_pain).
patient_symptom(ravi, headache).
patient_symptom(priya, sneezing).
patient_symptom(priya, runny_nose).
patient_symptom(priya, sore_throat).
patient_symptom(arun, fever).
patient_symptom(arun, chills).
patient_symptom(arun, sweating).
patient_symptom(arun, headache).
patient_symptom(divya, headache).
patient_symptom(divya, nausea).
patient_symptom(divya, sensitivity_to_light).
patient_symptom(kiran, high_fever).
patient_symptom(kiran, rash).
patient_symptom(kiran, joint_pain).
patient_symptom(kiran, headache).
patient_symptom(sam, cough).

% symptoms_of(Disease, ListOfSymptoms)
symptoms_of(flu, [fever, cough, body_pain, headache]).
symptoms_of(common_cold, [sneezing, runny_nose, sore_throat]).
symptoms_of(malaria, [fever, chills, sweating, headache]).
symptoms_of(dengue, [high_fever, rash, joint_pain, headache]).
symptoms_of(migraine, [headache, nausea, sensitivity_to_light]).

% advice(Disease, Advice)
advice(flu, 'Take rest, drink fluids and see a doctor if fever persists').
advice(common_cold, 'Take rest, drink warm fluids and steam inhalation').
advice(malaria, 'Consult a doctor immediately for a blood test').
advice(dengue, 'Consult a doctor immediately and drink plenty of fluids').
advice(migraine, 'Rest in a dark, quiet room and consult a doctor').

% ---------- Rules ----------
% A patient has a disease if ALL symptoms of the disease are present
diagnose(P, D) :-
    patient(P),
    symptoms_of(D, List),
    forall(member(S, List), patient_symptom(P, S)).
% Print the full report of a patient
```

```
report(P) :-
    (   \+ patient(P)
    ->  format("~w is not registered.~n", [P])
    ;   diagnose(P, D)
    ->  advice(D, A),
        format("Patient   : ~w~n", [P]),
        format("Diagnosis : ~w~n", [D]),
        format("Advice    : ~w~n", [A])
    ;   format("Patient   : ~w~n", [P]),
        format("Diagnosis : No matching disease. Please consult a doctor.~n")
    ).
```

OUTPUT:

```
$ swipl -q exp28.pl

?- diagnose(ravi, D).
D = flu.

?- diagnose(P, malaria).
P = arun.

?- diagnose(P, D).
P = ravi, D = flu ;
P = priya, D = common_cold ;
P = arun, D = malaria ;
P = divya, D = migraine ;
P = kiran, D = dengue.

?- advice(dengue, A).
A = 'Consult a doctor immediately and drink plenty of fluids'.

?- report(arun).
Patient   : arun
Diagnosis : malaria
Advice    : Consult a doctor immediately for a blood test
true.

?- report(sam).
Patient   : sam
Diagnosis : No matching disease. Please consult a doctor.
true.

?- report(john).
john is not registered.
true.

$
```

Result

The Medical Diagnosis program was implemented successfully using Prolog. The program compares the symptoms of a patient with the symptoms of each disease and reports the matching disease along with the advice.